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**B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2014.**

Third Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. State any four advantages of object oriented programming.
2. Write the properties of friend function.
3. What is destructor?
4. What is function overloading? Give example.
5. Explain how to create dynamic objects.
6. Write a small program explaining Virtual functions.

7. What are the various modes of file operation?
8. What is a class template? How to define it?
9. What are the advantages of UML?
10. What are the Patterns?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the major principles of object oriented programming with illustrations and neat sketches.

Or

12. Explain the decision making statements of C++.

UNIT II

13. Explain binary operator overloading with example program.

Or

14. Explain any two types of inheritance with example program.

UNIT III

15. What is the use of new and delete operators. Explain briefly overloading of the new and delete operator?

Or

16. Explain Virtual function with a program.

UNIT IV

17. Explain the file pointers and file manipulators.

Or

18. Explain Function Template with good example program.

UNIT V

19. Explain the interaction and state chart diagram of UML.

Or

20. Explain the concept of Interaction Modeling.

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B.Tech. DEGREE EXAMINATION, APRIL 2015.

Third Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

(2013-2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Outline the disadvantages of procedural programming.
2. Explain Lvalue and Rvalues.
3. State the usage of Constructor and Destructor.
4. Explain abstract class.
5. Write the function for malloc() and free().
6. Write sample C++ code to display the length of string.

UNIT III

7. Outline the different types of file modes.
8. List the type of errors encountered while reading (Or) writing file.

9. Explain swim lane and life lane in UML.

10. State the usage of adapter pattern design.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each unit.

All questions carry equal marks.

UNIT I

11. Illustrate various control structures in C++.

Or

12. Discuss various building blocks for object oriented programming language.

UNIT II

13. Illustrate binary operator overloading function with a sample example.

Or

14. Explain multiple inheritances with a sample example.

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15. Discuss different types of array declarations with a sample example.

Or

16. Design a program to print the output of various string handling methods.

UNIT IV

17. Develop a program to read and write array of integers into binary file.

Or

18. Discuss exception handling in C++.

UNIT V

19. Construct the UML usecase diagram to withdraw amount in ATM banking.

Or

20. Design Sequence and Collaboration diagram for issue and return of books in library.

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2017.

Third Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

(2013 – 14 Onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What are the advantages of OOP?
2. What is the use of the word friend keyword in C++?
3. Write some special characteristics of constructor.
4. List out the operators that cannot be overloaded.
5. List the two types of polymorphism? Give an example for each.
6. How are virtual functions declared in C++?
7. What are the three keywords used for exception handling mechanism?

8. Explain the usage of template in C++.

9. What do you know about UML?

10. What is interaction modelling?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the features of object oriented programming. Describe how each of these is implemented in C++.

Or

12. Explain how to overload a function with an example program in C++.

UNIT II

13. Explain the different types of Constructors that are available in C++.

Or

14. What is the significance of inheritance and explain the different types of inheritance with a programming example for each type.

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UNIT III

15. Explain the use of Virtual Functions with example program in C++.

Or

16. Explain how to concatenate two strings using example program in C++.

UNIT IV

17. Explain in detail about exception handling mechanism and write a program in C++ to illustrate the divide by zero exception.

Or

18. Explain the file handling techniques in C++.

UNIT V

19. Describe in detail: Object oriented analysis design methodology.

Or

20. Explain how to use the activity diagram interaction diagrams.

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2017.

Third Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

(2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Compare structured programming and object oriented programming.
2. Mention the use of array of objects.
3. Define copy constructors.
4. What is meant by virtual class?
5. How can we access private variables using pointers?
6. Why do we need dynamic objects?

UNIT III

15. Write a C++ program to find maximum among five numbers using array of pointers.
- Or

16. Write a C++ program that implements area of rectangle and triangle using abstract classes.

UNIT IV

17. Write a C++ program that illustrates class templates.
- Or

18. Explain exception handling in detail.

UNIT V

19. Explain various UML diagrams in detail.
- Or

20. Describe various methodology used for OOAD.
- _____

7. What is the need of templates?
8. List various file operation modes?
9. Where is activity diagrams used?
10. What are the various types of interaction diagrams?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Write a C++ program to find matrix inverse using functions.

Or

12. Write C++ programs that use both recursive and non-recursive functions.

- (a) Factorial of a given number
(b) GCD of two given integers.

UNIT II

13. Write a C++ program to overload unary minus operator.

Or

14. Write a C++ program that illustrates hybrid inheritance.

UNIT IV

Explain the steps of file operation with example.

Or

Write a program to create template – based stack.
Store float and integer in it.

UNIT V

Draw and explain the class diagram for a simple banking application.

Or

Explain interaction diagrams with an example.

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B.Tech. DEGREE EXAMINATION, APRIL/MAY

Third Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Write few concepts of object – oriented programming.

2. What is function overloading?

3. What is meant by copy constructor?

4. Describe the syntax of multiple inheritance. When do we use such an inheritance?

5. Give an example to create array of objects using pointers.

6. What is the need of virtual function?
7. What is command line argument? Give an example.
8. What are the advantages of using exceptions?
9. What is meant by activity diagram?
10. Define interaction modelling.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE question from each Unit.

UNIT I

11. Write a program to evaluate the following function

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

Or

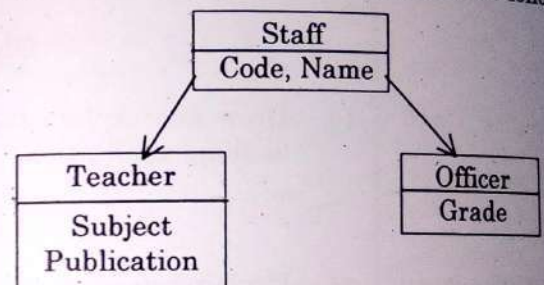
12. (a) What is a friend function. What are the merits and demerits of using friend function? (5)
- (b) Write a program to explain the inline function. (6)

UNIT II

13. Write a program to add two complex numbers using binary operator overloading concept.

Or

14. Write a C++ program to implement the following:



UNIT III

15. (a) Write a program to initialize the private numbers and display them without the use of member function.
- (b) Write a program to declare void pointers and access member variables using void pointers.

Or

16. (a) Write a program to create dynamic object.
- (b) Write a program to invoke function using scope access operator.

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B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2016.

Third Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

(2013 – 2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What is data hiding?
2. What is a friend function?
3. Define virtual class.
4. What is an abstract class?
5. What is meant by an array? Give an example.
6. State the use of new and delete operators.

7. Mention the use of templates.

8. What is a file mode?

9. What is UML?

10. State the use of class diagram.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, choosing ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the structure of the C++ program with an example.

Or

12. What is meant by function overloading? Explain in detail.

UNIT II

13. Discuss about dynamic initialization using constructors in detail.

Or

14. What is inheritance? Explain in detail.

UNIT III

15. Discuss about pointer to derived classes and base classes.

Or

16. Explain about declaring and initializing string objects in detail.

UNIT IV

17. Discuss command line arguments in detail.

Or

18. Explain about Exception handling in detail.

UNIT V

19. Discuss about interaction diagrams in detail.

Or

20. Explain about state chart diagrams in detail.

UNIT IV

17. Explain any two methodology concepts prescribed by Jacobson et al. (11)

Or

18. (a) Give a detailed note on Super-sub class relationship and a-part-of relationship? (5)
(b) Describe the UML class diagram with neat sketch. (6)

UNIT V

19. List out and explain the four testing strategies used in OODesign. (11)

Or

20. What do you mean by quality assurance test? Give its necessity and explain any two testing strategies. (11)

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2013.
Third Semester

Electronics and Communication Engineering

C++ AND OBJECT ORIENTED PROGRAMMING

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Compare objects and classes.
2. Object-Based Programming does not support inheritance and dynamic binding. True or False? Justify your answer.
3. Define operator overloading. List out the operators that cannot be overloaded.
4. Briefly explain the three class access specifiers.
5. Define multiple inheritances and how it helps reusability in OOSD.

6. What do you mean by Rapid Application Development?
7. What is a pattern?
8. Define static and dynamic model in UML.
9. Briefly explain the purpose of View layer interface.
10. Define OODesign and why do users find OOUI easier to use.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. (a) With the help of a program explain the arithmetic operations using pointers. (6)
- (b) Explain the characteristics of Object Oriented Languages. Give the applications of OOPs. (5)

Or

12. (a) Briefly explain the fundamental data types of C++. (5)
- (b) What do you mean by structured programming and mention its limitations? Give the benefits of object oriented programming. (6)

UNIT II

13. What is Inheritance? Explain the various types of Inheritance. Mention its advantages. (11)

Or

14. (a) Explain how Class Templates are used in C++. (5)
- (b) Explain Operator overloading with an example. (6)

UNIT III

15. Explain the reusability in object oriented programming using inheritance property with example. (11)

Or

16. (a) Briefly describe the system development process in object orientation. (5)
- (b) Why do we go for object oriented systems development? (3)
- (c) What are horizontal, vertical and domain prototype? (3)

UNIT IV

17. (a) Explain in detail about exception handling in C++ using sample code. (6)
(b) Write a C++ program using function template to find the product of two integer or floating point type of data. (5)

Or

18. Create a class Player with data members are playerid, playername, playergame and member functions are getplayer(), Searchplayer() and displayplayer(). Use file to store N employers information and perform those operation.

UNIT V

19. Derive the use case model for student course registration system in a University with proper interaction and activity diagram which include the following tasks Publication of Courses offered, student enrollment for courses, Attendance maintenance, Conduct of Exams and marks maintenance and Result publication.

Or

20. (a) Explain in detail about patterns with an example. (4)
(b) How an activity and state chart diagrams are used for modeling? (7)

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2015.

Third Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

(2013-2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Define Encapsulation and Data hiding.
2. State the advantages of inline functions in C++?
3. What is an abstract class?
4. What are the difference between default and parameterized constructors?
5. What is pure virtual function? Give an example.
6. How is a string object created?
7. What is Uncaught Exception? What is the causes of it?
8. What are the three standard template library adapters?

9. What do you know about use case model?
10. Write any two goodness criteria of object oriented design.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, ONE question from each unit.
All questions carry equal marks.

UNIT I

11. (a) Explain the basic concepts of Object oriented programming. (7)
- (b) How are pointers used to access the class members? Explain with an example. (4)

Or

12. (a) What are manipulators? Explain in detail about the various manipulators used for I/O operations. (7)
- (b) Explain in detail about friend function in C++ with an example. (4)

UNIT II

13. Create a class Vector to hold an array of integers and perform the following operations using constructor for dynamic allocation of array based on size given as argument
- (a) Create two vectors using constructor

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- (b) Add the vectors and return the resultant vector
- (c) Multiply all elements in the vector by a given number X and display the resultant vector.

Or

14. Write a C++ program to find the area of various 2D shapes such as square, rectangle, triangle, circle and ellipse using function overloading

UNIT III

15. (a) Explain in detail about the various run time casting in C++. (6)
- (b) Write a C++ program for calculating the area of rectangle and circle using run time polymorphism. (5)

Or

16. (a) Write a program in C++ to read two strings and perform the following string manipulation function (7)
- (i) Find the long string
- (ii) Compare the two strings
- (iii) Concatenate them.
- (b) Explain in detail about dynamic objects. How are they created? (4)

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